

CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics					
Course Code	STA2MN108 (P)					
Course Title	Statistics for critical thinking II					
Type of Course	Minor					
Semester	II					
Academic Level	100 - 199					
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours	

	4	3	-	2	75
Pre-requisites	Basic mathematical knowledge, familiarity with functions, graphs and basic equations.				
Course Summary	This course explores different ways to collect data, builds a foundation on probability, describes how to model a random experiment effectively using random variable and discusses some special distributions.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Design and develop questionnaires and schedules, ensuring reliability and validity through appropriate measures and techniques and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Quiz
CO2	Evaluate the reliability of data through various techniques and analyze data to help entrepreneurial decisions using critical thinking skills..	U	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Utilize random variables to model outcomes of random experiment.	U	C	Seminar Presentation / Group Tutorial Work
CO4	Define probability and analyze scenarios involving disjoint or mutually exclusive outcomes using probability concepts and tools.	U	C	Instructor-created exams / Home Assignments/ Instructor-created exams
CO5	Identify and describe key properties of common statistical distributions.	Ap	P	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Explore various probability distributions such as Bernoulli, binomial, Poisson, geometric, and negative binomial distributions, including understanding their	Ap	P	Viva Voce/ Instructor-created exams

	characteristics and differences using R software.			
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

COURSE CONTENT

Module		Content	Hours (45 +30)	Marks (70)
		Methods of Data Collection	9	10
1	1	Case study, Observation, Interview, Survey, Use of Secondary Data	3	
	2	Questionnaires and Schedules : Reliability and Validity of Questionnaire	3	
	3	Cleaning Data, Methods to Check reliability of Data.	3	
		Sections from References: Unit 1-3:		
		Probability	11	20
2	4	Defining probability, Disjoint or mutually exclusive outcomes, Probabilities when events are not disjoint, Venn-diagrams.	2	
	5	Probability distributions, Complement of an event, Independence.	2	
	6	Exploring probabilities with a contingency table, Marginal and joint probabilities.	1	
	7	Defining conditional probability, General multiplication rule.	2	
	8	Sum of conditional Probabilities, Independence considerations in conditional probability, Tree diagrams.	2	
	9	Bayes' Theorem and its applications.	2	
		Sections from References: Unit 4-9: 3.1,3.2 [Ref 2]		
		Continuous distributions	14	20
3	10	Sampling from a small population, without replacement, with replacement.	1	
	11	Random variable and its Expectation.	2	

	12	Variability in random variables.	2	
	13	Linear combinations of random variables, its Expectation and Variability in linear combinations of random variables.	2	
	14	Continuous distributions, From histograms to continuous distributions.	1	
	15	Probabilities from continuous distributions.	2	
	16	Normal distribution, standard normal distribution.	2	
	17	Standardizing with Z-scores, Finding tail areas, examples.	2	
		Sections from References: Unit 10-15: 3.3,3.4,3.5 [Ref 2] Unit 16-17: 4.1 [Ref 2]		
		Discrete distributions	11	20
4	18	Bernoulli distribution, binomial distribution,	2	
	19	Normal approximation to the binomial distribution,	1	
	20	Poisson distribution.	3	
	21	Geometric distribution.	2	
	22	Negative binomial distribution, Binomial vs Negative binomial distribution.	3	
		Sections from References: Unit 18-22: 4.2,4.3,4.4,4.5 [Ref 2]		
5	PRACTICUM		30	
	Do practice problems in R software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Obtain the probability distribution		
	2	Plot the probability distribution		
	3	Obtain the cumulative distribution function		
	4	Plot the cumulative distribution function		
	5	Calculation of Probabilities from binomial distribution		
	6	Calculation of Probabilities from binomial distribution		
	7	Calculation of Probabilities from binomial distribution		

	8	Fitting of Binomial distribution		
	Sections from References: Unit 1: 3.3 [Ref 5] Unit 2: 3.3 [Ref 5] Unit 3: 3.3 [Ref 5] Unit 4: 3.4 & 3.5 [Ref 5] Unit 5: 3.4 & 3.5 [Ref 5] Unit 6: 3.4 & 3.5 [Ref 5] Unit 7: 3.4 & 3.5 [Ref 5] Unit 8: 3.4 & 3.5 [Ref 5]			
	Books and References: 1. Asthana, H. S., & Bhushan, B. (2016). <i>Statistics for social sciences (with SPSS applications)</i> . PHI Learning Pvt. Ltd.. 2. Diez, D. M., Barr, C. D., & Cetinkaya-Rundel, M. (2019). <i>OpenIntro statistics</i> . Boston, MA, USA:: OpenIntro. 3. Aron, A., Coups, E. J., & Aron, E. N. (2013). <i>Statistics for the behavioral and social sciences: A brief course: Pearson new international edition</i> . Pearson Higher Ed. 4. Sirkin, R. M. (2006). <i>Statistics for the social sciences</i> . Sage. 5. Sudha G Purohith, Sharad D Core, Shailaja R Deshmukh, <i>Statistics Using R</i> (2015) 6. Mukherjee, S. P., Sinha, B. K., & Chattopadhyay, A. K. (2018). <i>Statistical methods in social science research</i> (pp. 29-37). Springer Singapore. 7. Gupta, S. C. and Kapoor, V. K. (2002). <i>Fundamentals of Mathematical Statistics</i> . , 11 th edition, Sulthan Chand, New Delhi. 8. Gupta, S. C. and Kapoor, V. K. (2007). <i>Fundamentals of applied Statistics</i> , Sultan Chand and Sons.			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO	2	2	-	-	-	3	1	2	-	-	-	3

1												
CO 2	-	-	3	-	-	2	2	1	-	-	3	-
CO 3	-	-	-	-	-	-	3	2	-	-	-	-
CO 4	-	3	-	3	2	-	2	2	3	-	-	-
CO 5	-	-	-	-	-	-	2	1	-	-	-	-
CO 6	-	-	3	-	-	2	1	2	-	3	-	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓			✓
CO 3	✓	✓		✓